

LESSON 2-1 • HOMEWORK PACKET • SHEET 1 OF 2**Statistical Questions**

Maryland Standard 6.SP.A.1 • Learning target – I can tell whether a question is statistical by deciding if the answers will vary.

REMEMBER

- A **statistical question** expects answers that **vary**. You collect **data** to answer it.
- A **non-statistical question** has one fixed answer, like a fact you can look up.
- A number answer alone does not make a question statistical. Ask: "Will the answers be different?"

WORKED EXAMPLE

Is "How many minutes does each student in Room 12 ride the bus?" statistical?

Step 1 Who answers? Every student in Room 12.

Step 2 Will the answers differ? Yes: 5 min, 12 min, 25 min ...

Step 3 The answers vary, so you need data.

Answer Statistical

PRACTICE

show your work • label units

1. Write **S** if the question is statistical or **N** if it is non-statistical.

Question	S or N
How many feet are in one yard?	
How many hours did each sixth grader sleep last night?	
What is the price of a bus ticket in Baltimore today?	
How many glasses of water does each person in my family drink a day?	

2. **Name Lengths.** Ms. Adeyemi asks each student, "How many letters are in your first name?" Five students answer 5, 4, 7, 5, and 6. Is her question statistical? Use the answers to explain.

☐ Statistical

☐ Not statistical

Because _____

3. **Survey Plan.** Is "How far does each player live from the field?" statistical?

☐ Statistical

☐ Not statistical

4. **Rewrite It.** Carlos asks, "How tall is the oak tree in front of our school?" Explain why this is not statistical. Then rewrite it as a statistical question.

5. **Your Turn.** Write a statistical question you could ask the people in your family. Then list three answers you might get.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Ask a family member one statistical question and one non-statistical question. Can they tell which is which?

LESSON 2-1 • HOMEWORK PACKET • SHEET 2 OF 2**Dot Plots and Shape**

Maryland Standard 6.SP.B.4 • Learning target – I can build a dot plot and describe the shape of the data it shows.

REMEMBER

- A **dot plot** shows one dot per data value.
- **Peak**: tallest stack. **Gap**: no dots. **Outlier**: far away.
- **Symmetric**: the sides match. **Skewed right/left**: the tail goes that way.

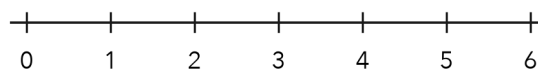
WORKED EXAMPLE

Pets owned: 0, 0, 1, 1, 1, 1, 2, 2, 3, 5. Name the shape.

Step 1 Stacks: 2 dots at 0, 4 at 1, 2 at 2, 1 at 3, 1 at 5.**Step 2** Peak at 1. The tail stretches right to 5.**Answer Skewed right****PRACTICE***show your work • label units*

- 1. Books Read.**
- Books read in May by 12 students:

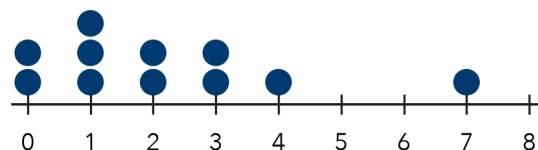
2, 3, 1, 3, 4, 2, 3, 5, 3, 2, 4, 4. Make a dot plot.

a. Peak: _____**b.** Shape: _____**Books Read in May**

- 2. Arcade Tickets.**
- Tickets won by 9 friends: 5, 3, 4, 5, 2, 5, 4, 6, 5. Picture the dot plot.

a. Dots at 5: _____**b.** Peak: _____**c.** Shape: _____

- 3. Soccer Goals.**
- The dot plot shows the goals a team scored in each game.

a. Games played: _____**b.** Most common: _____**c.** Where is the gap? _____**Goals Scored**

- 4. Find the Error.**
- Leah says the Soccer Goals plot is skewed left. Explain her error.

- 5. Quiz Scores.**
- 1 dot at 5, 1 at 6, 2 at 7, 4 at 8, 5 at 9, and 3 at 10. Name the shape.

☐ Symmetric☐ Skewed left☐ Skewed right**TEST PRACTICE***choose the best answer*

- 6.**
- Stacks of 1, 3, 5, 3, and 1 dots sit above 2, 3, 4, 5, and 6. Which word describes the shape?

A Symmetric**B** Skewed left**C** Skewed right**D** Has a gap

- 7.**
- A dot plot has 4 dots above 10, 6 above 11, 2 above 12, and 3 above 13. How many values?

A 4**B** 6**C** 13**D** 15**CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** Count the letters in each family member's first name. Make a dot plot and name its shape.